AJ MENDES

DATA SCIENCE | MACHINE LEARNING | CYBERSECURITY ANALYTICS

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PROFILE

Computer science student focused on reproducible data analysis, machine-learning evaluation, and secure systems. Experience designing controlled clustering experiments, building time-aware model pipelines, visualizing results, and translating technical findings for collaborators.

EDUCATION

Colorado School of Mines

B.S. Computer Science | Minors in Military Science and Quantum Engineering | Cyber Defense Education Certificate | Expected May 2027 | GPA: 3.5

RESEARCH EXPERIENCE

Undergraduate Data Science Researcher

The Data Mine, Purdue University | CrowdStrike project | Jan 2026 - May 2026

- Built Python evaluation tools for K-means clustering on labeled synthetic cybersecurity data, reporting accuracy, adjusted Rand index, and normalized mutual information across controlled experiments.
- Tested robustness across noise, cluster-count, and feature-variation sweeps; observed approximately 98% cluster-assignment accuracy in labeled tests with up to 20 clusters.
- Developed centroid-matching and drift-visualization workflows using PCA, UMAP, and t-SNE; documented and presented findings with Purdue, Data Mine of the Rockies, and CrowdStrike collaborators.

PROJECTS

MarketPulse AI

Python | pandas | scikit-learn | TensorFlow/Keras | Streamlit

- Built a Python ML research pipeline that engineers returns, volatility, moving-average, volume, RSI, and MACD features for seven equities against SPY.
- Compared logistic regression, random forest, and dense neural-network models with time-aware backtests, transaction costs, threshold sweeps, automated reports, tests, and a Streamlit dashboard.

Secure Chat Application

Python | RSA-OAEP | AES-256-GCM | TCP

- Built a Python TCP client-server chat application using RSA-2048/OAEP key exchange and AES-256-GCM authenticated encryption; the relay forwards ciphertext without access to plaintext.

ADDITIONAL EXPERIENCE

AI Research Contributor

Handshake AI | 2026 - Present

- Create and source-verify complex research tasks used to evaluate advanced AI systems; analyze model responses for factual accuracy, research quality, and failure patterns.

Electrical Control Panel Technician

Xpect Solutions, Longmont, CO | Jun 2025 - Aug 2025

- Assembled and wired industrial electrical control panels from schematics, wiring diagrams, and technical drawings.

Construction and General Labor

Formit Right Concrete, Longmont, CO | Summers 2020 - 2024

- Completed five summers of full-time concrete, site-preparation, material-handling, and landscaping work in changing jobsite conditions and early-shift schedules.

TECHNICAL SKILLS

Languages: Python, SQL, R, Java, C++, TypeScript/JavaScript
Data/ML: pandas, NumPy, scikit-learn, TensorFlow/Keras, PCA, t-SNE, clustering, backtesting
Tools: Git, Linux, Streamlit, Matplotlib

LEADERSHIP AND CREDENTIALS

Leadership: Army ROTC Cadet, Colorado School of Mines
Credential: Secret security clearance